Detection and Analysis of Monetary Policy Signals in Central Banks Minutes with TF-IDF, FinBERT, and ModernBERT

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Abstract

Central banks deliberately communicate interest rate policy decisions with neutral and opaque language to avoid premature market reactions. Efforts to identify forward-looking intent are thus a challenging area for NLP research,[1] particularly given that many extraneous economic factors influence rate changes. Building on prior work, we seek to determine whether semantic signals are sufficiently detectable using NLP classification on a dataset of 696 meeting minutes spanning 2000-2026 from the Federal Open Market Committee (FOMC), Bank of England (BoE), and Reserve Bank of Australia (RBA). We evaluate four model families in this study using logistic regression as a classifier: a majority-class baseline, TF-IDF as a lexical baseline, chunked FinBERT[5] for domain-specific semantic representation, and frozen ModernBERT for full document context. Documents are split temporally (2000-2019 train, 2020-2026 test) to determine whether predictive signal exists. Results confirm that predictive signal does exist and relies on semantic rather than lexical patterns. TF-IDF performs nearly identically to the majority-class baseline on FOMC meeting minutes text (F1 macro 0.405 and 0.372 respectively), while ModernBERT shows strongest predictive performance (F1 macro 0.690, + 0.285 over baseline), with 81% recall achieved in the change class (only missing 3/16 rate changes). Across all models, regime shifts with historically unprecedented policy actions were the most challenging to predict due to novel language relative to training data.

1 - Introduction

Central banks are the cornerstone of the global financial system. When the Federal Reserve Open Market Committee (FOMC), the Bank of England (BoE), or the Reserve Bank of Australia (RBA) periodically adjust interest rates (typically in small 25 basis point increments), the downstream impact on borrowing costs, asset prices, and economic growth potential are substantial. Committee members thus have to carefully weigh many economic factors including inflation, employment, and stability in their analysis. Each institution documents their process through meeting minutes released weeks after policy decisions, providing the public with insight into their conclusions.

In order to properly manage market expectations and avoid reactions, the language used in meeting minutes documents is intentionally opaque and neutral.[1] Furthermore, studies have shown that FOMC text in particular is challenging to analyze with NLP methods due to its inherent objectivity and complexity.[2] This may suggest that any potential signal in forward guidance is buried underneath subtle semantic patterns rather than surface-level word choice.

Previous NLP work has primarily focused on cross-examining central bank text with immediate market reactions[3] and sentiment analysis[4], rather than predicting signal on future policy decisions. Kim et al. (2023)[2] also demonstrates that models specifically adapted for this domain, such as FinBERT[5], offer modest advantages in performance over general language models missing domain-specific fine tuning. Our work expands on this through the following goals. We first want to evaluate the ability to predict future policy decisions from three major central banks (FOMC, BoE, RBA) as a binary outcome (rate

held vs. adjusted) using prior period meeting minutes text. We compare the performance between a lexical model baseline (TF-IDF), a domain specific semantic model with 512 token limit (FinBERT), and a general context semantic model with 8,192 token limit (ModernBERT).[6] We also seek to identify any linguistic patterns driving predictions and whether policy signal is universal or institution-specific.

2 - Related Work

Applying NLP to central bank text is a challenging task, with Blinder et al. (2008)[1] establishing that lexical methods struggle due to intentionally opaque language. However, more recent studies that leverage increasingly complex semantic approaches have found some success. Doh et al. (2020)[4] used a text-based semantic approach to find subtle forward looking cues in FOMC communication that extended beyond what the macroeconomic states provided alone. However, their primary focus was to capture immediate market sentiment rather than predicting future outcomes. Additionally, Handlan (2022)[3] applies neural network approaches to FOMC text to predict immediate market reactions, further demonstrating the predictive potential of central bank language. However, neither study explicitly focuses on predicting the binary outcome of rate policy changes in the following decision.

Related to our work, Kim et al. (2023)[2] provide a benchmark evaluation of FinBERT, VADER, and GPT-4 models in FOMC minute sentiment classification tasks. One major finding was that models with domain-specific fine tuning, like FinBERT, can modestly outperform general purpose models. This provided guidance into our model selection, with FinBERT as a domain-specific choice, unfortunately limited by token length of 512. Given that FOMC minutes text average 7,000 words, a chunking approach would be necessary with FinBERT. To complement FinBERT, we implemented the general purpose ModernBERT with token length of 8,192 to assess an approach without required chunking or truncation. Aruoba and Drechsel (2022)[7] supports our use of ModernBERT and longer context length by demonstrating that FOMC documents contain distributed policy cues from hundreds of economic concepts throughout.

To our knowledge, however, no study has specifically examined whether signal is sufficient to classify forward-looking decisions, nor attempt to identify whether signal generalizes across different central bank committees.

3 - Methods

3.1 Dataset

Text from 696 total meeting minutes were combined from three central bank institutions: FOMC (236, 2000-2026), BoE (253, 2000-2026), and RBA (207, 2006-2026). FOMC minutes text was obtained from the vtasca HuggingFace dataset. No pre-existing dataset could be found that has consolidated meeting minutes for BoE and RBA, and each had to be scraped using a custom python script. BoE minutes were scraped by iteratively converting URLs to PDF endpoints with naming conventions that evolved over time. RBA minutes were more straightforward to obtain, with two URL date formats (YYYY-MM-DD and DDMMYYYY) required for before 2015 and after 2015 respectively.

Binary labels were assigned historically based on the effective policy rate change after each meeting. Rate changes were flagged as *change* and constant rates were flagged as *hold*. FOMC historical rates prior to

2008 were backfilled using the DFEDTAR series, while rates beyond 2008 were calculated by taking the midpoint of the DFEDTARU (upper bound) and DFEDTARL (lower bound) series. All three of these sources were obtained from CSV extracts downloaded from federalreserve.gov. Both BoE and RBA historical rates were obtained and backfilled from daily frequency exports: Bank of England historical CSV and RBA's A2 statistical table.

To ensure accuracy, we validated the combined and labeled dataset with multiple known minutes dates and cuts spanning the COVID era and prior easing cycles. Several duplicate entries were found and removed, and one source document was missing (FOMC March 2009). See **Table 1** below for a summary:

Table 1: Dataset Summary

Bank	Meetings	Change	Hold	Change %	Date Range	Median Words
FOMC	236	68	168	28.8%	2000–2026	7,339
BoE	253	52	201	20.6%	2000–2026	4,707
RBA	207	48	159	23.2%	2006–2026	3,142
Total	*696*	*168*	*528*	*24.1%*	*2000–2026*	—

3.2 Train/Test Split

A temporal split was applied to the dataset. When training the combined models, minutes text prior to 2020 were used for training (535 total), and 2020 onwards formed the test set (161 total). See **Table 2** below for a summary of the sample size contributions from each central bank:

Table 2: Train/Test Split Summary

Bank	Train (2000–2019)	Test (2020–2026)	Total
FOMC	186	50	236
BoE	204	49	253
RBA	145	62	207
Total	*535*	*161*	*696*

3.3 Models

Four model approaches in total were used: **majority-class baseline** (assumes *hold*), **TF-IDF** (lexical baseline), **FinBERT** (semantic, domain-specific, chunking required), and **ModernBERT** (semantic, general-purpose, full context). Each of the latter three models feature a two-stage architecture with a downstream logistic regression (LR) model used to convert either a sparse term frequency vector (TF-IDF) or dimensional embedding (FinBERT, ModernBERT) to binary classifier output. Both FinBERT and ModernBERT are *frozen*, in other words training weights are not updated during training, due to small training data sample size.

Majority-class baseline: Model performance floor, which predicts *hold* for every meeting outcome.

TF-IDF + LR: Uses a weighted bag of bigrams to represent each minutes document, with year tokens specifically removed to avoid memorization or data leakage. Used as the lexical baseline to test whether surface-level word frequency contains any signal to influence future policy decisions.

FinBERT (Frozen, Chunked) + LR: BERT encoder that is pre-trained on financial text to provide domain-specific context. Limited to 512 tokens, which requires chunking of minutes text due to typical length of 3,000 to 7,000 words. Chunks were split using a 64-token overlap and maximum of 20 per document. Chunk embeddings are then converted to a single dimensional embedding per document via mean-pooling, which is then fed to an LR model.

ModernBERT (Frozen) + LR: ModernBERT encoder used as a general-purpose feature extractor. Each full minutes document is tokenized and passed through the model with a limit of 8,192 tokens. The output yields a dimensional embedding that is then fed to an LR model.

3.4 Evaluation

Due to class imbalance (~75% *hold*, ~25% *change*), accuracy was not used as a primary metric to evaluate model performance. **Macro F1** is a more reliable performance indicator because it equally penalizes the majority and minority class for inaccuracy. Majority hold baseline, for example, would yield a combined accuracy of 67%, but F1 macro of 0.401. We also report **Precision** (fraction of predicted changes that were actually correct) and **Recall** (fraction of actual changes detected) for the *change* class specifically.

4 Results and Discussion

4.1 Main Results

Table 3: F1 Macro, Precision/Recall by Model and Central Bank (Test Set, 2020-2026)

Model	F1 Macro				Precision / Recall (change)	
	FOMC	BoE	RBA	Combined	Precision	Recall
Majority Class	0.405	0.372	0.421	0.401	0.000	0.000
TF-IDF + LR	0.405	0.372	0.621	0.592	0.000	0.000
FinBERT Chunked + LR	0.612	0.403	0.578	0.625	0.444	0.750
ModernBERT + LR	**0.690**	**0.424**	**0.669**	**0.627**	**0.520**	**0.812**

Table 3 summarizes the main findings of each model run. Most importantly, ModernBERT + LR consistently outperformed all other models, with the greatest improvement over the majority class baseline shown in FOMC minutes text (+0.285). While 0.690 is not objectively high in absolute terms, this nonetheless suggests genuine predictive signal within the contents of the FOMC minutes text, which is dissected in more detail in Section 4.2. Similar results are shown for both the Combined (+.226) and RBA (+.248) model runs.

FinBERT also performed well despite token length limitations and necessary chunking strategy (F1 macro 0.612 on FOMC). The domain-specific pre-training provides strong signal, capturing policy-related

semantic context, but chunking limits the ability to learn longer range patterns where ModernBERT succeeds. The comparison nonetheless addresses a tradeoff between context length and domain adaptation, with the former more essential for this specific task.

Interestingly, all model approaches performed poorly on BoE minutes text, including ModernBERT (F1 macro 0.424). This is likely attributed to the small representation of rate changes in the 2000-2019 training set (15.7%) versus the 2020-2026 test set (40.8%), primarily driven by the 2022-2023 hiking cycle. The models were thus unprepared for the linguistic patterns of an aggressive hiking regime that is lacking in training data. We cannot confirm that poor policy signal alone was responsible for bad performance.

4.2 Semantic vs. Lexical Signal in FOMC Minutes

The stark contrast between TF-IDF and ModernBERT performance on FOMC minutes text strongly suggests that policy signal is coded semantically, and not lexically. TF-IDF on FOMC performs identically to the majority-class baseline (F1 macro 0.405), indicating a complete lack of useful lexical signal. This is confirmed through vocabulary analysis with type-token ratio (TTR), which measures lexical diversity across change and hold FOMC meetings (0.441 and 0.444 respectively). ModernBERT's self-attention mechanism is better equipped to capture the underlying semantic meaning despite no changes in vocabulary.

4.3 Error Analysis and Regime Transitions

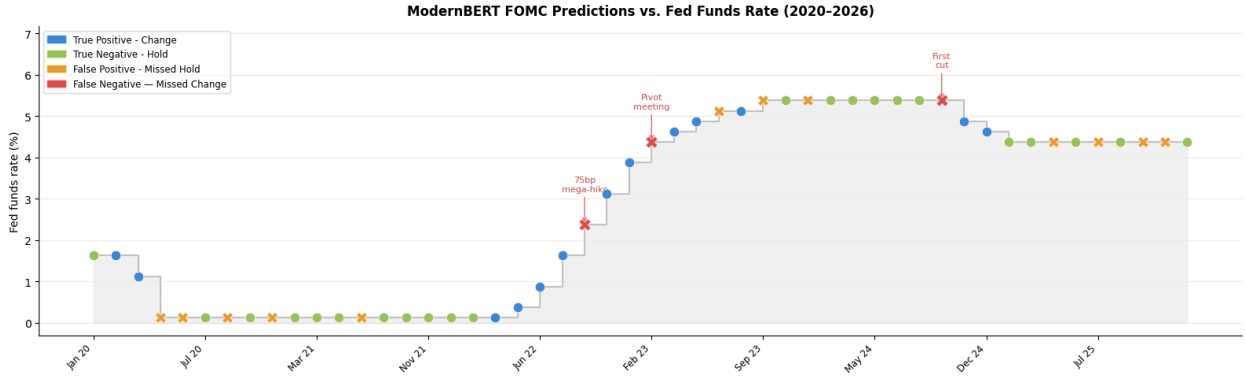
ModernBERT correctly predicts 35/50 FOMC policy decisions in the test set (70% accuracy, F1 Macro 0.690) while achieving 81% recall on the change class (13/16 rate changes). Interestingly, the three false negative FOMC minutes text each correspond to atypical moments in the test set:

- **September, 2022**: 75 basis point hike, the largest single hike in over 40 years
- **February, 2023**: mid-cycle pivot, Fed shifting from aggressive rate hikes to a pause
- **September, 2024**: the first rate cut of the easing cycle, start of a new policy regime

Linguistic analysis of these meetings reveal a few interesting patterns. The term "disinflation" appears at a rate of 0.130 per 1,000 words but is otherwise absent from all correctly predicted change meetings. "Restrictive" and "cumulative" are found roughly twice as frequently in false negatives (1.252 vs. 0.721 per 1,000 words, 0.475 vs. 0.206 per 1,000 words respectively). These terms are components of new policy language from the 2022-2023 pivot ("sufficiently restrictive" and "cumulative tightening effects") that were not present in the training documents. Introduction of new vocabulary with no historical precedent likely influenced the mislabeling of these events.

False positives also show interesting patterns, which were primarily clustered around two distinct periods. Unprecedented crisis language from the Covid era hold meetings (April-December, 2020) and the more recent extended hold period (2024-2025) are absent from training data. ModernBERT likely conflates these extended hold periods because they are largely absent from the training data. This pattern is illustrated in **Figure 1**, which clearly shows the concentration of false negatives at policy inflection points and false positives at extended hold periods, where performance is otherwise strong.

Figure 1: ModernBERT FOMC Rate Predictions vs. Federal Funds Rate (2020-2026)



5 - Conclusion

This study evaluates whether meeting minutes from three major central banks (FOMC, BoE, and RBA) contain sufficient linguistic signal to classify future policy decisions. Despite the intentional opacity of their language, we find that genuine predictive signal exists, but is only accessible in models capable of capturing full-document semantic context.

Lexical approaches fail almost entirely, with TF-IDF finding no incremental improvement in predictive signal over the majority class baseline with FOMC nor BoE. Only RBA minutes show a meaningful result (F1 macro 0.621), likely attributed to data leakage with explicit decision language embedded in RBA documents. BoE results are consistently weak across all models, with the 2022-2023 hiking cycle dominating the test period but absent from training.

Semantic signal is most clearly demonstrated on FOMC text, with each BERT transformer model outperforming the lexical and majority-class baseline. Chunked FinBERT demonstrates that domain-specific fine tuning provides a meaningful signal (F1 macro 0.612, +0.207 over baseline). However, ModernBERT's larger 8,192 token context window achieves a stronger result (F1 macro 0.690, +0.285 over baseline), with 81% recall and detection of 13 out of 16 Fed rate changes. In comparing FinBERT and ModernBERT, we find that full-document context is more essential than domain adaptation for this specific task.

Error analysis of ModernBERT's FOMC misses indicates that the model struggles at specific moments. False negatives are present at regime transitions, and false positives at extended rate hold periods. Novel language is introduced that are absent from training data, including "disinflation" and "sufficiently restrictive" that are seen at much higher frequency in missed documents. The ability for the model to learn regime-invariant policy signal is a challenge for time-dependent financial NLP tasks.

Future work will ideally explore models that combine the domain adaptation of FinBERT and larger context window of ModernBERT for better performance, the logical next step for this task.

Limitations

Several limitations are worth noting for this study. Sample size of publicly available central bank meeting minutes text is limited, with only 535 training examples across all three institutions. This limitation prevented the possibility of transformer fine-tuning, and we had to rely on pre-existing frozen BERT models. Minor labeling noise also exists due to BoE meeting dates approximated to the first of each month, a consequence of scraping constraints. Lastly, the test set for all central banks contains historically unprecedented macroeconomic policy regimes due to the Covid pandemic and high lasting inflation leading to extended policy hold periods. Results are also limited to English-speaking minutes text and may not generalize well to other central bank contexts.

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AI Disclosure

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